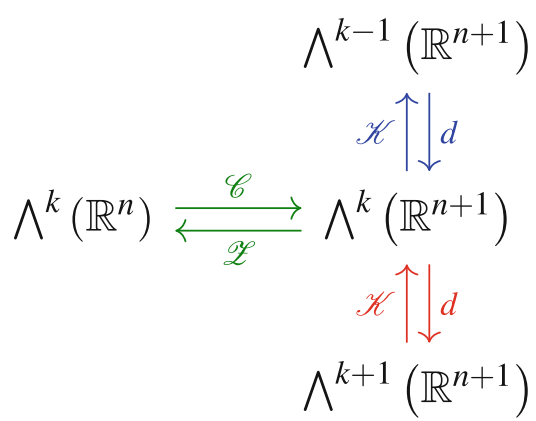
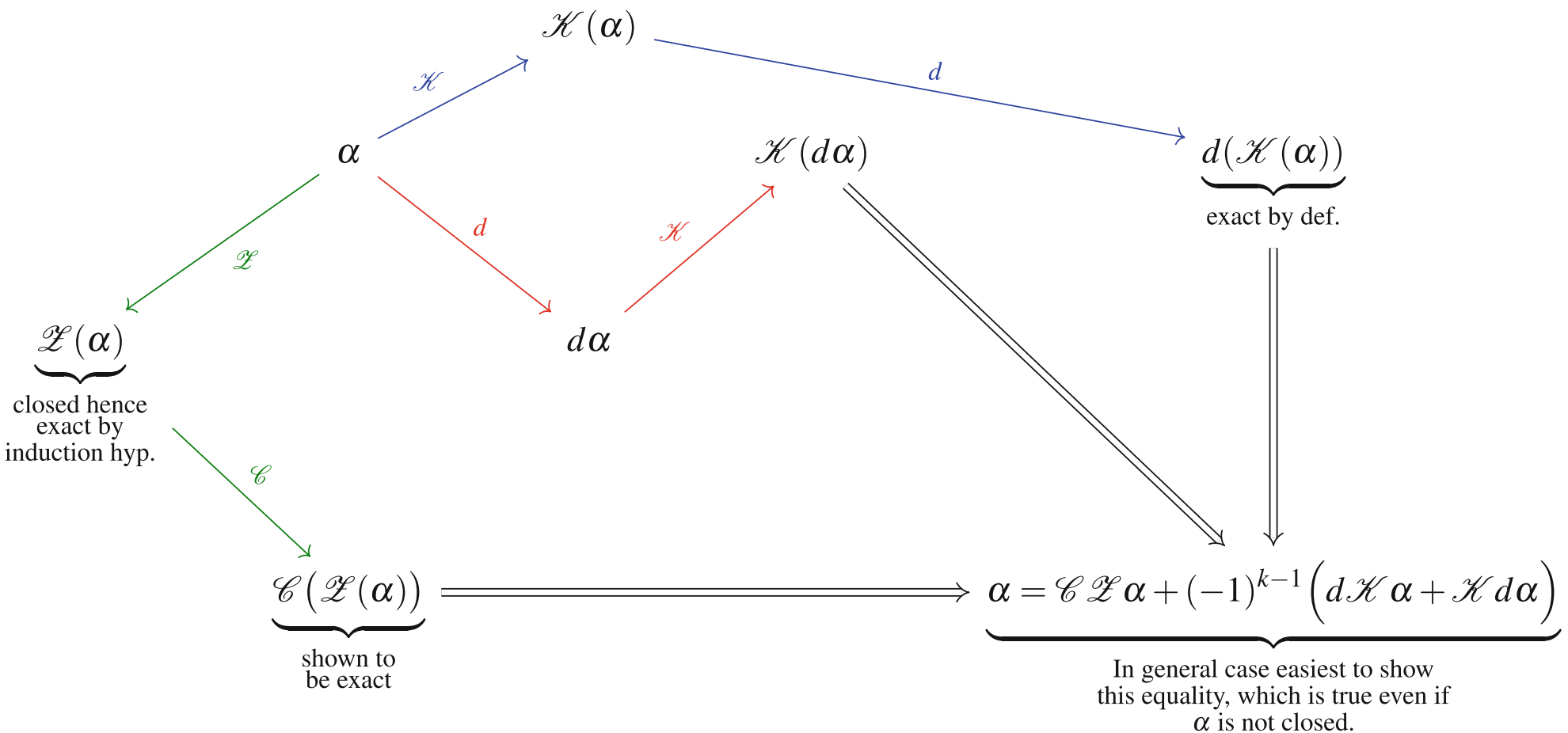

$\bigwedge^{k-1}(\mathbb{R}^{n+1})$
$\mathscr{K}$
d
$\bigwedge^{k}(\mathbb{R}^{n})$
$\mathscr{C}$
$\mathscr{Z}$
$\bigwedge^{k}(\mathbb{R}^{n+1})$
$\mathscr{K}$
d
$\bigwedge^{k+1}(\mathbb{R}^{n+1})$


$\mathscr{K}(\alpha)$
$\mathscr{K}$
d
α
$\mathscr{K}(d\alpha)$
$d(\mathscr{K}(\alpha))$
exact by def.
d
$\mathscr{K}$
$\mathscr{Z}$
$\mathscr{Z}(\alpha)$
closed hence exact by induction hyp.
$d\alpha$
$\mathscr{C}$
$\mathscr{C}(\mathscr{Z}(\alpha))$
shown to be exact
$\alpha = \mathscr{C}\mathscr{Z}\alpha + (-1)^{k-1}\left(d\mathscr{K}\alpha + \mathscr{K}d\alpha\right)$
In general case easiest to show this equality, which is true even if α is not closed.